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2 Personalized AI assistants could either promote or hinder the
3 emergence of collective intelligence

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5 Jaeseung Kim¹ Sang Woo Park^{1,2,*}

6 **1** School of Biological Sciences, Seoul National University, Seoul, Korea

7 **2** Institute for Data Innovation in Science, Seoul National University, Seoul, Korea

8 *Corresponding author: sangwoopark@snu.ac.kr

9 **Abstract**

10 Advances in artificial intelligence (AI) have enabled the wide use of personalized
11 assistance for cognitive tasks, which enhanced individual performance on various
12 fields. However, the impact of AI assistance on collective intelligence remains poorly
13 understood. AI assistance may undermine the emergence of collective intelligence by
14 reducing the individual diversity or introducing structured biases into a population.
15 Here, we develop a human-AI collective prediction model mediated by social learning
16 based on incentive structure by extending the existing works on collective intelligence.
17 Our model uncovers that emergence of collective intelligence is depend on how human
18 and AI interact. When AI works as if it is omniscient and gives assistance regardless
19 of users' interests, collective intelligence will be promoted. However, when AI works
20 as a chat-bot and gives assistance only based on users' interests, collective intelligence
21 could be hindered due to diversity loss and overconfidence on AI. We showed that,
22 to promote collective intelligence, penalizing or incentivizing AI usage is not reliable.
23 Instead, elaborating more diversity-aware incentives could be the key to promote and
24 maintain the collective intelligence.

Introduction

Collective intelligence (CI) describes a phenomenon in which a group’s ability to solve problems, make decisions, or adapt to changing environments exceeds each member’s ability [1, 2, 3, 4, 5]. This phenomenon, also referred to as wisdom of crowds, has thus served as a central mechanism underlying the success of diverse organizations, ranging from animal groups [6, 7, 8] and human societies [2, 9] to machine-learning systems [10, 11]. Understanding the conditions that promote and maintain CI therefore remains a fundamental question across social sciences [12], evolutionary biology [13], and computer science [14]. Addressing this question has become particularly timely as artificial intelligence (AI) has begun to reshape how individuals acquire information, make decisions, and contribute to collective outcomes [15].

The recent emergence of generative AIs and large language models (LLMs) has changed the way individuals engage in various cognitive tasks, ranging from simple everyday tasks to more complex tasks, including scientific discovery [16] and creative writing [17]. In particular, while the use of personal AI models, such as ChatGPT, has been rapidly increasing [18], the potential long-term impact of personal AI services on human society, especially the CI, remains uncertain. Some argue that AI can promote CI by enhancing individual expertise [19], and opinion aggregation [20]. Conversely, empirical evidence indicate that the use of AIs can limit the diversity of opinions [17, 21, 22], a key requirement for the emergence of CI [23, 24, 25].

Our ability to predict the long-term impact of personal AI services on human CI is hampered by the apparent discrepancy in the incentives that shape AI uses. On one hand, many industrial sectors are pushing towards increased AI usages, where the number of “tokens” used is often perceived as a proxy for productivity [26, 27]. On the other hand, many academic institutions appear to discourage, and even penalize, AI use, reflecting concerns about plagiarism, misinformation, and limited independence [28, 29]. Such discrepancy raises a question about whether directly incentivizing or penalizing AI usage will benefit or harm human CI. These observations further highlight the importance of indentifying a robust incentive structure that can promote CI in the presence of AI, which can retain the benefits of AIs while preventing potential risk such as preserving the diversity of individual opinions.

To address these questions, we extend previous approaches for modeling CI, where each individual makes independent predictions for the outcome from partial observations of underlying causal factors [12, 13, 30]. We allow each individual to independently consult with personal AIs and revise their predictions accordingly. Across different assumptions about the AI model’s ability and underlying incentive structures, we investigate the robustness of human CIs.

Main

Model

We begin by extending previously established models of CI in which a group of players participates in a prediction task, with each player making an individual prediction based on partial observations of the underlying causal factors and the group forming a collective prediction by aggregating these individual predictions [30, 12, 13]. The prediction target (Y) is modeled as a linear combination of M causal factors (x_i), with fixed coefficients (α_i): $Y = \sum_{i=1}^M \alpha_i x_i + \alpha_0$, where each factor is drawn from independent normal distribution. Among M different factors, an individual (A) makes prediction by the multiplication of single factor of interest (x_{e_A}) and their own belief (c_A): $Y_A = c_A x_{e_A}$. Here, we assume each individual can independently consult AI and revise their prediction based on their reliance on AI (β_A): $Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A$, and the revised prediction is aggregated to form collective prediction.

To model AI, we consider a scenario where the group of players is given a novel prediction task, whereas the AI has an ability to perform prediction based on causal factors because it already been trained on similar tasks. Every time a player consults AI, it will make predictions (Y_A^{AI}) based on its internal parameters, which can include factor-specific bias (b_i) and stochastic error (ϵ). We then consider two extreme types of AI response, which differ in the amount of information AI uses when making predictions. At one extreme, the AI operates with complete knowledge of all causal factors, using them regardless of the user’s interest: $Y_A^{AI} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon$. At the other extreme, AI can work like a chatbot, in which AI predicts the outcome based only on the user’s interest: $Y_A^{AI} = \alpha_{e_A} x_{e_A} + b_{e_A} + \epsilon$. Hereafter, we refer to these two types as ‘Omniscient AI’ and ‘Chatbot AI’, respectively.

Each generation consists of multiple round of predictions, and each player is rewarded by a payoff according to the incentive structure. For simplicity, we assume that a sufficiently large number of prediction rounds occurs within each generation, allowing us to use the expected payoff rather than computing the payoff for each round. We consider two incentive structures, “feedback” and “niche-expert” [13], which have been shown to promote near-perfect collective accuracy in the absence of AI (except averaging aggregation with niche-expert structure). The feedback structure gives higher payoff to individuals who can push the collective prediction toward the true target, regardless of their individual predictive accuracy. In contrast, the niche-expert structure gives higher payoff to individuals who have both better predictive accuracy and a minority interest.

Through time, the group evolves by imitating the peers who receive higher payoffs. Specifically, after each generation, two individuals are randomly sampled, and one imitates all three strategies (factor of interest, belief, reliance on AI) of the other with a probability determined by the difference between their expected payoffs. Individual predictions are aggregated into a collective prediction ($\hat{Y}$) by simple averaging when

103 using Omniscient AI and by clustering when using Chatbot AI. Collective accuracy
 104 is defined as the reduction in prediction error achieved by the collective prediction
 105 relative to a baseline prediction (α_0). Hereafter, we refer to the near-perfect collective
 106 accuracy as accurate CI.

107 This model now allows us to explore the impact of AI on CI depending on the
 108 mode of AI and the incentive structure. Here, we begin by comparing the emergence
 109 of CI with and without AI and subsequently investigate a better incentive structure
 110 that can promote CI in the presence of AI.

111 **AI can either promote or prevent the emergence of CI**

112 To understand the impact of AI, we first compare collective dynamics with and
 113 without AI. We find that, except when Omniscient AI is used under the feedback
 114 structure, a group with AI fails to achieve accurate CI (Fig. 2). However, the effect
 115 of AI differs depending on the mode of AI.

116 When Omniscient AI assists players, the group becomes equally or more accurate
 117 than the group without AI. Under the feedback structure, a group achieves accurate
 118 CI regardless of the assistance of AI (Fig. 2A). However, under the niche-expert
 119 structure, the group without AI is no better than the baseline prediction. Although
 120 Omniscient AI cannot help players achieve accurate CI, the group’s prediction with
 121 AI is more accurate than both that of the group without AI and the AI itself (Fig.
 122 2B). These results are robust across different levels of AI accuracy and can be shown
 123 mathematically by assuming a sufficiently large group of players (Fig. S2).

124 When Chatbot AI assists players, the group becomes less accurate than the group
 125 without AI. Although both incentive structures can achieve accurate CI without AI,
 126 the assistance of Chatbot AI hinders group performance and prevents accurate CI
 127 from emerging (Fig. 2C, D). This occurs through different mechanisms depending
 128 on the incentive structure. Under the feedback structure, Chatbot AI leads to a loss
 129 of interest diversity, which eventually makes it impossible for the group to account
 130 for variation in some causal factors. This leads to an increase in collective variance
 131 and stochastic decrease in collective accuracy (Fig. S1). The loss of interest diversity
 132 may arise even without AI after sufficiently many generations in a finite population.
 133 However, Chatbot AI makes the group more vulnerable to the diversity loss. Un-
 134 der the niche-expert structure, the group becomes overly reliant on AI, causing the
 135 collective prediction to inherit the AI’s bias (Fig. S1). This leads to an increase in
 136 collective bias and a decrease in collective accuracy. The niche-expert structure may
 137 also drive reliance on AI toward zero, depending on the relationship between the
 138 true coefficient and AI bias (Fig. S2). In this case, the group completely excludes
 139 AI assistance and builds CI on its own, which substantially delays the emergence of
 140 accurate CI.

141 To investigate whether the group can retain its predictive ability when AI as-
 142 sistance suddenly becomes unavailable at a given generation, we measure counter-
 143 factual collective accuracy. This measure follows the same evolutionary trajectory

as the simulation with AI, but accuracy is computed using the players’ unrevised predictions. We find that counterfactual collective accuracy is always lower than the collective accuracy of groups, regardless of the AI mode or incentive structure.

In summary, AI assistance may promote CI when it is omniscient, whereas AI operating with limited information may prevent the emergence of accurate CI. These findings underscore the need for incentive structures that can mitigate the potential risks associated with AI assistance.

Directly incentivizing or penalizing AI use can additionally harm CI

Given the strong influence of AI on collective outcomes, a straightforward intervention is to directly incentivize or penalize the use of AI. To investigate whether such direct incentives or penalties help emergence of CI, we modify the two incentive structures by adding a payoff term proportional to individual reliance on AI:

$$\text{Feedback/Niche-expert} + \lambda\beta_A \quad (1)$$

where λ represent the strength of incentive ($\lambda > 0$) or penalty ($\lambda < 0$).

Directly incentivizing or penalizing AI use further reduce the collective accuracy (Fig. 3C, E, G), except when Omniscient AI is used under the feedback structure (Fig. 3A). This is because incentivizing or penalizing AI usage pushes the group to always ($\beta = 1$) or never ($\beta = 0$) rely on AI, respectively (Fig. 3B, D, F, H). In some cases, when AI bias is relatively large and the penalty is weak, penalizing AI use can achieve higher collective accuracy than the original incentive structure. However, in most cases, penalizing AI use imposes strong selection against AI reliance, pushing the group toward lower stationary collective accuracy. Although the group with Chatbot AI under feedback structure shows stochastic breakdown of CI which doesn’t result in stationary collective accuracy, incentivizing or penalizing AI use further strengthens the breakdown (Fig. 3E). This result suggests that simply incentivizing or penalizing the AI use may exacerbate rather than resolving the problem of AI.

Balanced incentives mitigate AI-induced failures of CI

Neither the original incentive structures nor simple interventions targeting AI use are sufficient to promote CI. What incentive structure, then, would work in the presence of AI? Focusing on the Chatbot AI, in which proposed incentive structures fail to promote accurate CI, we investigate whether a better structure can overcome the risks introduced by AI. Here, we propose a “balanced” structure, defined as the simple average of the feedback and niche expert incentives.

Core idea behind the balanced structure is to maintain interest diversity while preventing the collective bias of the group. The feedback structure is prone to interest diversity loss, whereas can strongly prevent collective bias by incentivizing players whose predictions lie opposite to the collective prediction. By contrast, the

181 niche-expert structure can maintain interest diversity because it directly incentivizes
182 minority interests. However it cannot prevent collective bias because, within the
183 same factor of interest, individuals are rewarded only according to their individual
184 predictive performance, which homogenize the predictions.

185 By integrating both structure, the balanced structure can promote accurate CI by
186 maintaining both interest diversity and preventing collective bias (Fig. 4, S3). Com-
187 pare to the feedback and niche-expert, players are both evenly distributed through
188 factors, and beliefs are close to the real coefficient (Fig. 4). Moreover, counterfac-
189 tual collective accuracy maintained high compare to both feedback and niche-expert.
190 These patterns are driven by the emergent property of the balanced structure, in
191 which group increases its reliance on AI as the AI becomes more accurate, demon-
192 strating appropriately calibrated AI use.

193 Discussion

194 In this paper, we study how AI and incentive structures affect the emergence of CI.
195 Our model suggests that the effect of AI depends on the type of AI used by the
196 group. AI that can incorporate all relevant causal factors may improve group perfor-
197 mance, whereas AI that relies on users to gather information about the causal factors
198 may impair it. By examining different variants of incentive structures, we further
199 show that direct interventions targeting AI use generally harm group performance.
200 Instead, we suggest balanced structure which can promote CI while also generating
201 appropriately calibrated AI use as an emergent property.

202 Although Omniscient AI can produce group performance that is equal to or bet-
203 ter than that achieved without AI, such AI may be difficult to realize in practice
204 because it needs to be able to observe causal factors independently. Achieving this
205 capability would require hardware, such as sensors, as well as autonomous data ac-
206 quisition capabilities. For this reason, most AI services currently available are better
207 represented by the Chatbot AI in our model, which highlights the potential risks of
208 AI use for CI.

209 In our model, individuals revise their predictions based on the AI’s assistance. It
210 is therefore unsurprising that AI assistance reduces the diversity of individual pre-
211 dictions, a pattern that has also been observed empirically. However, a reduction in
212 prediction diversity does not necessarily lead to the collapse of CI. The principal risk
213 arises when the incentive structure fails to guide the group toward an appropriate
214 set of strategies. The feedback structure leads to the breakdown of CI because it
215 recruits more players to more important factors, which have larger coefficients. This
216 characteristic creates a risk of interest-diversity loss in finite populations, a situa-
217 tion in which no player remains interested in certain factors because most players
218 shift toward relatively important factors. Without AI, this occurs when initial in-
219 dividual predictions are not sufficiently diverse or when there is no source of novel
220 ideas, such as mutation. Revising individual predictions with AI assistance exac-

221 erbates this problem by pulling individuals’ predictions toward the AI’s prediction,
222 thereby reducing prediction diversity even under the same belief distribution. The
223 niche-expert structure hinders CI because it favors extreme levels of reliance on AI.
224 Because individuals within each interest group are affected only by their own perfor-
225 mance, incentives based on individual performance reach their maximum at either
226 full reliance on AI or no reliance at all. This extreme reliance results in suboptimal
227 collective accuracy or substantially delays the emergence of CI.

228 Because AI already generates substantial benefits as well as costs, organizations
229 increasingly seek to improve collective performance by regulating AI use. Such in-
230 terventions may take direct forms, including AI-use policies or token-based leader-
231 boards, or may operate indirectly through other incentives, such as productivity
232 rewards or the social stigma associated with AI use. However, our model suggests
233 that directly incentivizing or penalizing AI use is unlikely to induce appropriately
234 calibrated reliance on AI or to promote CI. Incentivizing AI use could lead to overre-
235 liance, causing groups to propagate AI biases, whereas penalizing AI use could lead
236 to underreliance, preventing groups from using AI even when it performs well in the
237 relevant domain. Our result instead shows that appropriate AI use and accurate CI
238 can be promoted by focusing on the classic properties of groups that have been em-
239 phasized for decades—maintaining diversity and preventing collective bias—rather
240 than on AI use itself.

241 The consistently limited counterfactual collective accuracy observed across AI
242 types and incentive structures suggests that the presence of AI can degrade hu-
243 man group’s own performance, consistent with empirical evidence. This may reduce
244 collective performance when AI assistance becomes unavailable because of resource
245 constraints, system failure, or model collapse, forcing the group to rebuild CI through
246 further adaptation. The higher counterfactual collective accuracy achieved under the
247 balanced structure indicates that an appropriate incentive structure can mitigate the
248 risk of CI breakdown following the loss of AI assistance.

249 Our model and analysis have several limitations. First, the model considers only
250 social learning among humans and treats the influence of AI as unidirectional. In
251 reality, humans can learn from AI, while AI systems can also be further trained
252 on human-generated data. Such bidirectional adaptation may further reduce group
253 performance by increasing the similarity between human and AI predictions and
254 thereby weakening their complementarity. Empirical studies of human–AI collabora-
255 tion likewise suggest that stronger correlations between human and AI predictions
256 can diminish the benefits of collaboration. Second, we model AI errors using factor-
257 specific additive biases and random noise. Previous studies of AI bias and error
258 suggest that additive biases may arise from skewed training data, or systematic cal-
259 ibration. However, AI errors may also exhibit more complex structures, including
260 biased estimates of factor effects and correlations across prediction errors. Further
261 empirical research is therefore needed to characterize the structure of AI bias and
262 error, together with modeling approaches that can incorporate these more complex
263 patterns. Finally, to retain the tractability of established CI models, we assume that

the prediction target is a linear combination of causal factors, which does not hold for many real-world tasks. In particular, many machine learning methods used in AI systems are capable of capturing nonlinear relationships. Future research should therefore investigate how nonlinear relationships between causal factors and prediction targets affect the interaction between AI assistance and the emergence of CI.

Methods

To model CI, we start with an abstract model that assumes the prediction target is the result of multiple causal factors [13]. Specifically, the prediction target Y is modeled as a linear combination of M independent factors $(x_1, x_2, \dots, x_m)$ and $M + 1$ coefficients $(\alpha_0, \alpha_1, \dots, \alpha_m)$, where α_0 denotes the global offset (Fig. 1A). The M factors are randomly drawn from normal distributions with mean X_i and standard deviation σ_i in each generation.

Among a population of N individuals, each individual can observe a single factor and make a prediction based on the observed value and their own belief. In particular, individual A 's prediction is modeled as:

$$Y_A = c_A x_{e_A} \quad (2)$$

where c_A represents individual A 's personal belief, e_A represents the factor that A observes (interest), and x_{e_A} represents the realized value of A 's interest. Individuals can also choose not to observe any factor, which represents stubborn individuals who make predictions only based on their own belief. For notational convenience, we assume that stubborn individuals have interest 0, and let x_0 always be 1.

Next, we model AI as a prediction system that has learned the true coefficients with bias and error (Fig. 1B). For each factor $i \in [0, M]$, AI knows the exact coefficient α_i and has a fixed bias b_i , resulting in a prediction of $\alpha_i x_i + b_i$. In terms of human-AI interaction, we model two contrasting modes of AI: the ‘‘AI knows all’’ model and the ‘‘AI answers question’’ model. In the AI knows all model, AI acts as an omniscient system and gives an answer regardless of the individual's interest, accounting for all factors related to the task. Thus, AI predicts as follows:

$$Y_A^{AI} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon \quad (3)$$

In contrast, the AI answers question model acts as a chatbot and gives an answer only based on the individual's interest. Thus, given the individual's interest, e_A , AI predicts as follows:

$$Y_A^{AI} = \alpha_{e_A} x_{e_A} + b_{e_A} + \epsilon \quad (4)$$

In both models, $\epsilon \sim N(0, \sigma_e^2)$ represents random error arising from the stochastic generative process of the AI model.

AI works as a personalized assistant for each individual, such that each individual receives assistance by incorporating the AI’s prediction. However, even with the same opportunity to receive AI assistance, reliance on AI can vary substantially across individuals. To model this, we assume that the revised prediction is a weighted sum of the individual’s own prediction and the AI’s prediction by introducing AI belief β_A as a weighting coefficient. The revised prediction of individual A can be written as:

$$Y_A^{revised} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \quad (5)$$

The revised predictions of individuals are aggregated by two different mechanisms, depending on the mode of AI. In the AI knows all model, even if an individual can observe only a single factor, their revised prediction contains information about all factors through interaction with AI. Therefore, we use simple averaging of revised predictions as the collective prediction. In contrast, in the AI answers question model, each individual gives a prediction based only on a single factor even after revising their prediction with AI assistance. Therefore, we use clustering aggregation to obtain the collective prediction, which is performed by clustering individuals based on interest and summing the average values of each cluster.

For each generation, each individual is rewarded with a payoff based on the given incentive structure (Fig. 1C). We adopt two incentive structures, “feedback” and “niche expert,” which were proposed by Wang et al. [13] and promote collective intelligence when prediction is performed without AI. The feedback incentive gives higher payoff to individuals who can push the collective prediction toward the true value, regardless of their individual performance. Given individual A ’s revised prediction $Y_A^{revised}$ and the collective prediction $\hat{Y}^{revised}$, it can be expressed as:

$$\pi_A = Y_A^{revised} (Y - \hat{Y}^{revised}) \quad (6)$$

The niche expert incentive gives higher payoff to individuals who have both better individual performance and a minority interest. Specifically, the niche expert payoff is expressed as the product of the proportion of individuals who observe the same factor as individual A , ρ_{e_A} , and individual accuracy. Given individual A ’s revised prediction $Y_A^{revised}$, it can be expressed as:

$$\pi_A = -\rho_{e_A} (Y_A^{revised} - Y)^2 \quad (7)$$

These two incentive structures were shown to achieve nearly perfect collective intelligence through social learning (except for the niche expert incentive with simple averaging aggregation) by promoting individual expertise while maintaining collective diversity. Therefore, we use these incentive structures as a baseline to understand how AI affects the emergence of collective intelligence. Moreover, we modify the two baseline incentives in a later section to understand the consequences of direct incentives and penalties for AI usage.

Throughout the simulation, individuals engage in social learning by imitating other individuals who receive higher payoffs. We assume that prediction events occur more frequently than social learning events within the population, such that individual payoff can be computed as an expected payoff. The imitation process is performed for at most one pair in each generation. By sampling two random individuals, A and B , imitation is performed with probability:

$$\frac{1}{1 + \exp(s(\mathbb{E}[\pi_A] - \mathbb{E}[\pi_B]))} \quad (8)$$

where s denotes the strength of social learning. When imitation occurs, A imitates all three strategies of B : interest, belief, and AI belief.

To evaluate and analyze the dynamics of collective prediction, we define six summary metrics: collective accuracy, collective bias, collective variance, human accuracy, interest diversity, and median AI belief. These metrics capture the accuracy of the revised collective prediction, its bias–variance decomposition, the predictive performance without AI, the diversity of individual interests, and median AI belief in the population (Table 1).

Table 1: Evaluation metrics used to analyze collective prediction dynamics.

Metric	Definition	Interpretation
Collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{revised}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the revised collective prediction relative to the baseline prediction α_0 .
Collective bias	$\left(\mathbb{E}[Y - \hat{Y}^{\text{revised}}]\right)^2$	Squared systematic deviation of the revised collective prediction from the true value.
Collective variance	$\text{Var}(Y - \hat{Y}^{\text{revised}})$	Variability of the revised collective prediction error across prediction events.
Counterfactual collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the human-only collective prediction relative to the baseline prediction α_0 .
Interest diversity	$\exp\left(-\sum_{i=0}^M \rho_i \ln \rho_i\right)$	Effective number of interests represented in the population.
Median reliance on AI	$\text{median}_{A \in N}(\beta_A)$	Median value of reliance on AI β_A in the population.

345 Data and code availability

346 All data are stored in a publicly available GitHub repository ([https://github.com/](https://github.com/parklab-snu/collective-intelligence-with-AI)
347 [parklab-snu/collective-intelligence-with-AI](https://github.com/parklab-snu/collective-intelligence-with-AI)).

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435 Contributions

436 Jaeseung Kim: Conceptualization, Formal Analysis, Investigation, Methodology,
437 Software, Visualization, Writing - Original Draft Preparation, Writing - Review &
438 Editing. Sang Woo Park: Conceptualization, Investigation, Methodology, Supervi-
439 sion, Writing - Original Draft Preparation, Writing - Review & Editing.

440 Competing Interests

441 The authors declare no competing interests.

442 Figures

Figure 1: **Observed changes in RSV outbreak dynamics in Japan.** (A) Relative RSV cases across 47 prefectures in Japan between 2013 and 2024. Relative cases, representing the relative magnitude of RSV outbreaks in each prefecture compared to RSV outbreak sizes before the COVID-19 pandemic, are calculated by dividing the raw cases by the maximum value before the COVID-19 pandemic. The red arrow indicates when the shift in seasonality occurred. (B) Estimates of center of gravity (i.e., the mean timing of an epidemic) across 46 prefectures, excluding Okinawa. (C) Estimates of the epidemic trough (i.e., the minimum number of cases in a season divided by the population size) across 47 prefectures. In panels B and C, the center (horizontal line), lower bounds, and upper bounds of the box plot correspond to median, lower quartile (25th percentile), and upper quartile (75th percentile), respectively. The whiskers indicate the range of values that extend up to 1.5 times the interquartile range beyond the lower and upper quartiles. Outliers, which fall outside this range, are plotted individually. (D) Time series of RSV cases across 5 major islands.

Figure 2: **Summary of SIRS model fits to RSV outbreaks across major islands in Japan.** (A) Comparisons of observed cases (points) across the five major islands and fitted epidemic trajectories (red lines). (B) Estimated periodic seasonal transmission rates. (C) Relationship between the estimated periodic seasonal transmission rates and mean specific humidity. Points represent seasonal transmission rate estimates across 52 weeks versus average humidity across 2013–2020. Blue lines and regions represent the corresponding locally estimated scatterplot smoothing (LOESS) estimates and corresponding 95% confidence intervals ($n=52$). Red lines and regions represent the corresponding quadratic regression fits and corresponding 95% confidence intervals ($n=52$). (D) Estimated relative changes in transmission, capturing the impact of NPI measures. (E) Estimated proportion of the susceptible pool. In panels B, D, and E, lines represent the estimated median of the posterior distribution ($n=8000$ posterior samples). In panels B, D, and E, shaded regions represent the 95% credible intervals from the posterior distribution ($n=8000$ posterior samples).

Figure 3: **An increase in the susceptible pool explains sudden changes in seasonality.** (A) Predicted effects of the proportion of infected $i(0)$ and susceptible $S(0)$ at the beginning of season on center of gravity. Points represent the estimated values for $i(0)$ and $s(0)$ between 2013 and 2019, showing the last two digits of a given year. The white vertical dashed line represents the $i(0)$ value used for simulating epidemic dynamics in panel B. (B) Changes in epidemic trajectories caused by an increase in the susceptible proportion at the beginning of season for a fixed value of $i(0)$. (C–F) Comparisons of interpolated transmission rates used for simulating the SIRS model, corresponding to each corner in Panel G. Black lines represent the transmission rates used for simulations. Gray lines represent the estimated transmission rates the Honshu island as a visual reference. (C) The estimated transmission rates for the Honshu island. (D) The resulting transmission rates for the Honshu island with equal amplitude as the Kyushu island. (E) The resulting transmission rates for the Kyushu island with equal amplitude as the Honshu island. (F) The estimated transmission rates for the Kyushu island. (G) Differences in peak epidemic timing when we increase the the susceptible proportion at the beginning of season from 7.8% to 10.5% using transmission patterns that interpolate the estimates for Honshu and Kyushu islands.

Figure 4: **Increase in the susceptible pool following the launch of the Comprehensive Support System for Children and Childcare in Japan.** (A) Direct comparisons between the estimates of susceptible proportion at the beginning of each season in each island (black) and the number of children attending childcare facilities in Japan (red). Points represent median from $n=8000$ posterior samples. Shaded regions represent the 95% credible interval in our estimates ($n=8000$ posterior samples). (B) Correlations between the estimates of susceptible proportion at the beginning of each season in each island and the number of children attending childcare facilities in Japan. Points represent median from $n=8000$ posterior samples. Error bars represent the 95% credible interval in our estimates ($n=8000$ posterior samples). Red lines and shaded regions represent the best fitting linear regression and the corresponding 95% confidence intervals.